

AuraCart Global

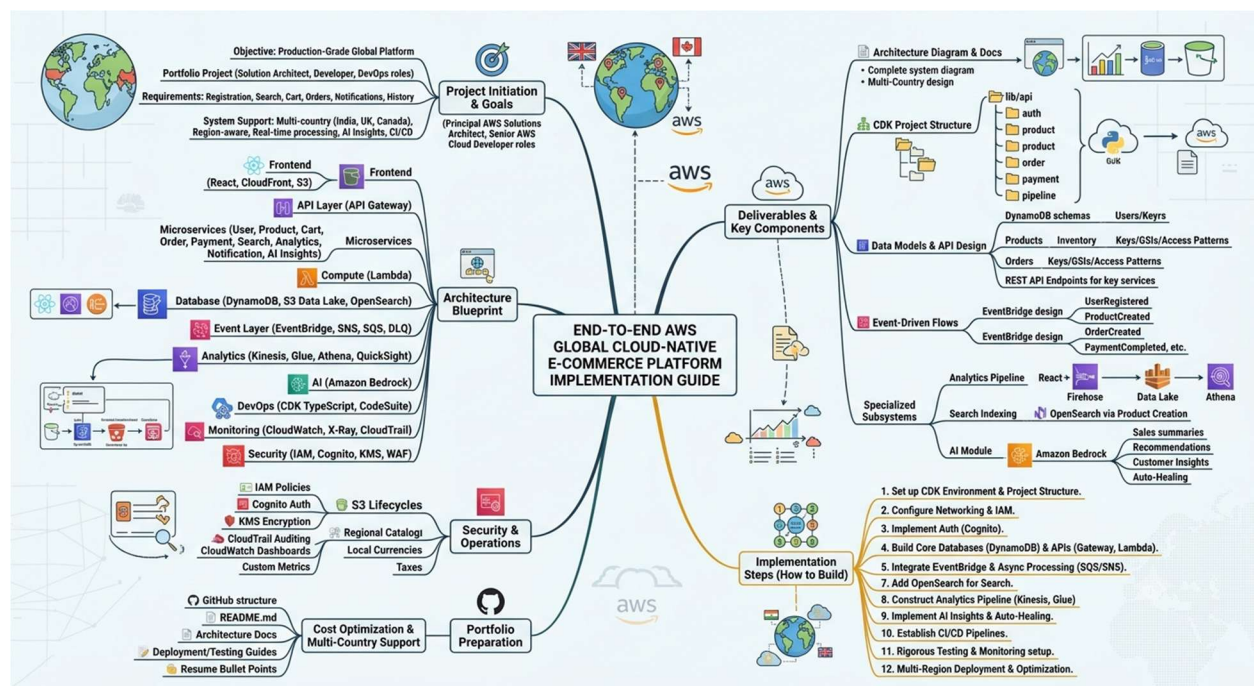
Building a Global Ecommerce Application on AWS

GlobalCommerce360 is a sophisticated, cloud-native e-commerce platform built on AWS, designed to serve customers across multiple regions (India, UK, Canada) with enterprise-grade scalability, security, and performance. The platform demonstrates advanced cloud architecture patterns including microservices, event-driven design, real-time analytics, and AI-powered insights.

Technical Architecture Overview

The system is built entirely on AWS, leveraging a serverless and managed-service philosophy to minimize operational overhead while maximizing scale, durability, and speed.

Global Ecommerce Website



End-to-End Data Flow

- Auth: User → CloudFront → S3 → API Gateway → User Service → Cognito → DynamoDB. *Result:* Event Bridge triggers SNS emails & Analytics.
- Search: User → API Gateway → Search Service → OpenSearch. *Result:* Fetches details and stock levels from DynamoDB.
- Cart: User → API Gateway → Cart Service → DynamoDB (with TTL). *Result:* Event Bridge tracks behavior; TTL clears old carts.
- Order: User → API Gateway → Order Service → DynamoDB. *Result:* Event Bridge handles payment gateway, drops inventory stock, and sends notifications.
- Analytics: Clicks → Kinesis → Firehose → S3. *Result:* Glue & Athena process the data for dashboards.

6. AI: Data → Bedrock AI. *Result:* Outputs recommendations, sales forecasts, and BI reports.

Core AWS Services & Business Value

Service	Purpose	Business Value
AWS Lambda	Serverless compute for microservices	60% cost reduction, infinite scalability
API Gateway	Unified API management and security	Single entry point, rate limiting, monitoring
DynamoDB	NoSQL database for high-performance operations	Sub-millisecond latency, auto-scaling
Aurora Serverless	Relational database with auto-scaling	ACID compliance, cost optimization
EventBridge	Event-driven architecture orchestration	Loose coupling, system resilience
SNS / SQS	Asynchronous messaging and notifications	Reliable delivery, fault tolerance
S3	Static website hosting and data lake	99.999999999% durability, cost-effective
CloudFront	Global content delivery network	50% faster load times, global reach
Route53	DNS management and traffic routing	100% uptime SLA, intelligent routing
Cognito	User authentication and authorization	Secure, scalable identity management
Kinesis	Real-time data streaming	Immediate insights, real-time decisions
Glue	ETL and data catalog management	Automated data preparation, governance

Service	Purpose	Business Value
Athena	Serverless SQL queries on data lake	Pay-per-query, no infrastructure management
QuickSight	Business intelligence dashboards	Data-driven decisions, visual insights
Bedrock	AI/ML models for recommendations	25% increase in conversion rates
OpenSearch / ES	Full-text search and analytics engine	Faceted search, auto-complete, enhanced discovery
CloudWatch	Monitoring, logging, and alerting	Proactive issue resolution, performance tuning
X-Ray	Distributed tracing and debugging	Faster troubleshooting, bottleneck insights
CloudTrail	API auditing and compliance	Security compliance, comprehensive audit trails
KMS	Encryption key management	Data protection at rest, regulatory compliance
WAF	Web application firewall	DDoS protection, advanced security filtering

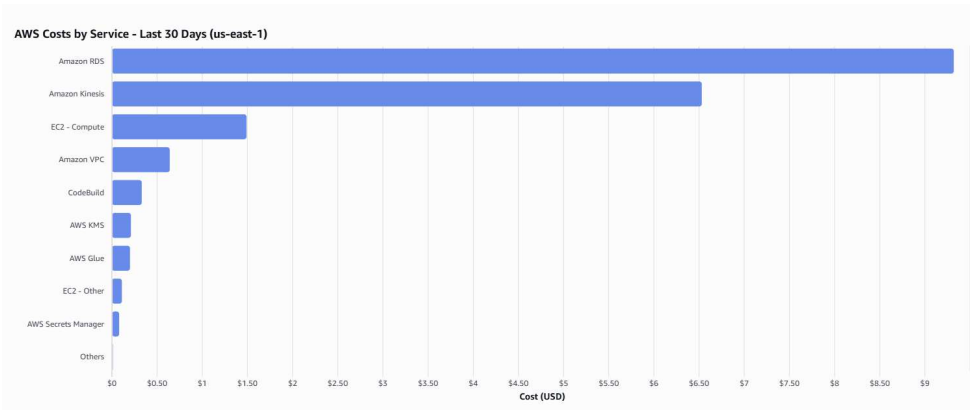
Versatile Application Fields

- **E-commerce & Retail:** Powers B2C/B2B marketplaces (Amazon/Shopify style), multi-vendor networks, and subscription models with recurring billing.
- **Digital Services:** Ideal backend for SaaS apps, media streaming/content delivery, developer API marketplaces, and Fintech/digital wallets.
- **Enterprise Applications:** Scales efficiently for Supply Chain logistics, CRM systems, ERP suites, and corporate HR management platforms.
- **Industry Solutions:** Tailored for Healthcare (telemedicine), Education (LMS), Real Estate (property management), and Travel (booking engines).

The Future Horizon Roadmap

- **1. Amazon SageMaker Integration:** Swapping out static evaluation rules for live Machine Learning inference endpoints via Lambda hooks.
- **2. Streaming Analytics:** Integrating Kinesis Data Analytics for sliding-window anomaly checks before the data hits cold storage.
- **3. Global Footprint:** Activating DynamoDB Global Tables and Amazon CloudFront to replicate microsecond data lookups across international borders.
- **4. Enterprise Guardrails:** Rolling out AWS GuardDuty and Security Hub alongside multi-tenant isolation layers to lock down compliance standards.
- **5. Next-Gen Commerce:** Plugging in AWS IoT Core and Alexa Skills to shift the platform from standard web screens to voice and smart-device shopping.
- **6. Edge Innovation:** Deploying application code directly to AWS Wavelength to tap into ultra-fast 5G networks and reduce latency to absolute zero.

Budget: My Project Budget and Cost Estimation



AWS DynamoDB: Tables

DynamoDB > Tables

DynamoDB

Dashboard

Tables

Explore items

PartiQL editor

Backups

Exports to S3

Imports from S3

Integrations

Reserved capacity

Settings

DAX

Clusters

Subnet groups

Parameter groups

Events

Tables (10)

Find tables

Filter by tag

Filter by tag value

June 27, 2026, 12:54 (UTC+5:30)

Actions

Delete

Create table

Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity mode
dev-cart	Active	userid (S)	cartid (S)	0	0	Off	☆	On-demand
dev-inventory	Active	warehouseid (S)	productid (S)	1	0	Off	☆	On-demand
dev-orders	Active	orderid (S)	itemType (S)	2	0	Off	☆	On-demand
dev-payments	Active	paymentid (S)	-	2	0	Off	☆	On-demand
dev-products	Active	productid (S)	-	2	0	Off	☆	On-demand
dev-users	Active	userid (S)	-	2	0	Off	☆	On-demand
ecommerce-inventory	Active	productid (S)	-	0	0	Off	☆	On-demand
ecommerce-orders	Active	orderid (S)	-	0	0	Off	☆	On-demand
ecommerce-products	Active	productid (S)	-	0	0	Off	☆	On-demand
ecommerce-users	Active	userid (S)	-	0	0	Off	☆	On-demand

DynamoDB > **Tables** > **dev-products**

Your feedback is an important part of helping us provide a better customer experience. Take this short survey to let us know how we're doing.

Tables (10) **dev-products** Last updated: June 27, 2026, 12:35 (UTC+5:30) Actions Explore table items

Filter by tag: Any tag key Filter by tag value: Any tag value Find tables

Get live item count

When you choose "Start scan," you will perform a DynamoDB scan to determine the most-recent item count. This scan might consume additional table read capacity units.

⚠ It is not recommended to perform this action on very large tables or tables that serve critical production traffic. You can pause the action at any time to avoid consuming extra read capacity.

Item count: 3 Scan status: Complete Last updated: June 27, 2026 12:36:01

Scan again Cancel

Additional info

Read/write capacity Info Edit

The read/write capacity mode controls how you are charged for read and write throughput and how you manage capacity.

DynamoDB > **Tables** > **ecommerce-inventory**

Partition key: productId (String) Sort key: Point-in-time recovery (PITR) info Capacity mode: On-demand Item count: 0 Table status: Active Table size: 0 bytes

Alarms: No active alarms Average item size: 0 bytes Amazon Resource Name (ARN): arn:aws:dynamodb:us-east-1:759945100268:table/ecommerce-inventory

Additional info

Read/write capacity Info Edit

The read/write capacity mode controls how you are charged for read and write throughput and how you manage capacity.

Capacity mode: On-demand

Maximum read request units: Maximum write request units

Auto scaling activities (0) Recent events of automatic scaling. Learn more

Find events

Start time	End time	Target	Capacity unit	Description	Status
No auto scaling activities found					

There are no auto scaling activities for the table or its global secondary indexes.

Warm throughput Info Edit

Prepare your table for planned peak events, without impacting your application performance or availability. Learn more about Amazon DynamoDB pricing

Name	Status	Type	Read units per second	Write operations per second
ecommerce-inventory	Active	Table	12,000	4,000

AWS API Gateway: Resources and Stage

API Gateway > **APIs** > **Resources - ecom-api (4kijfdpxrl)**

API actions Deploy API

Resources

Create resource

- /
 - /analytics
 - GET
 - OPTIONS
 - /(reportType)
 - GET
 - OPTIONS
 - /inventory
 - GET
 - OPTIONS
 - /productId
 - GET
 - OPTIONS
 - /orders
 - GET
 - OPTIONS

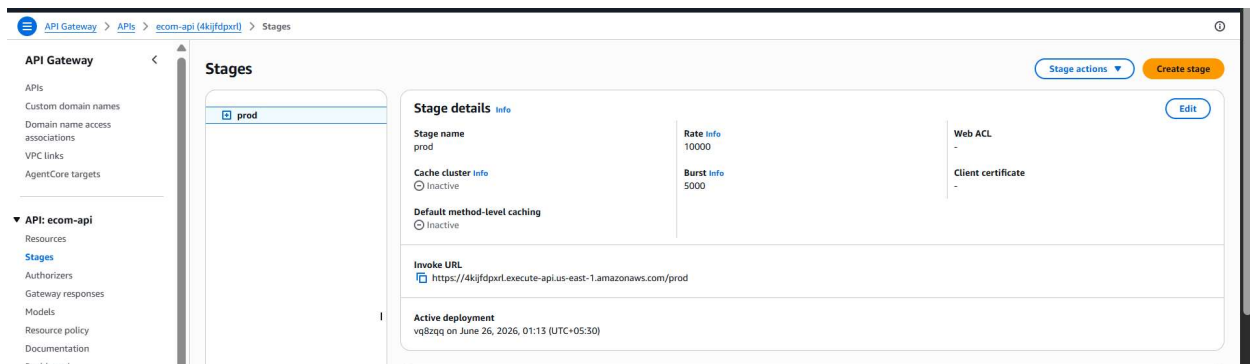
/analytics - GET - Method execution Update documentation Delete

ARN: arn:aws:execute-api:us-east-1:759945100268:4kijfdpxrl/*/GET/analytics Resource ID: v1r8j0

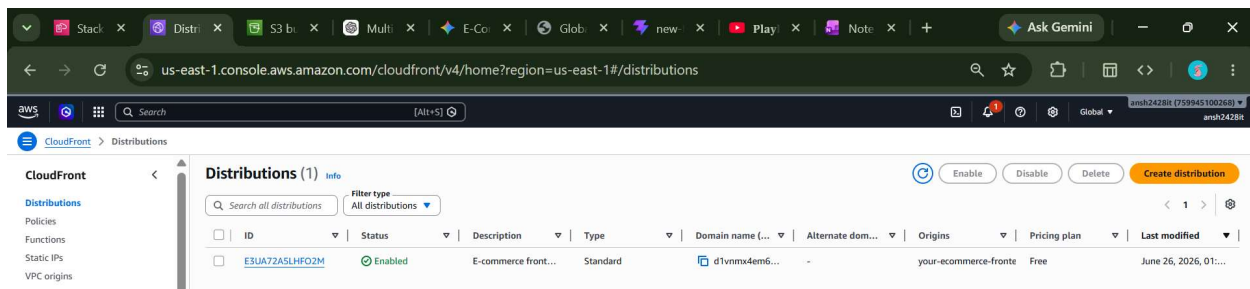
Client → Method request → Integration request → Integration response (Proxy integration) → Method response

Method request settings Edit

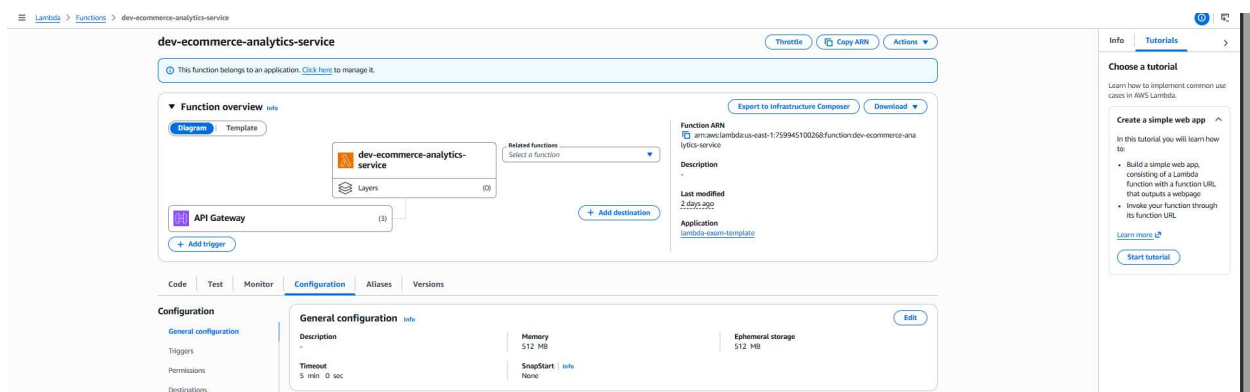
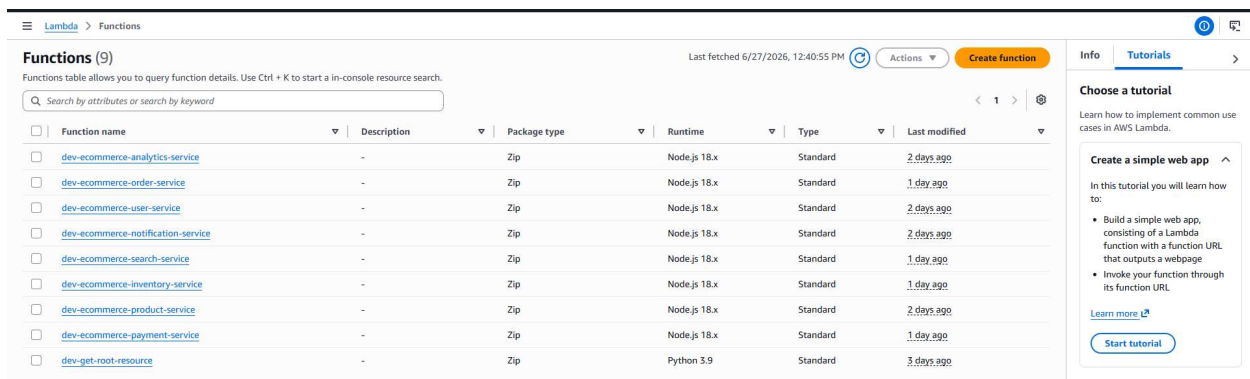
| Authorization | API key required |
|-------------------|------------------------------------|
| NONE | False |
| Request validator | SDK operation name |
| None | Generated based on method and path |



AWS CloudFront: Distributions



AWS Lambda: Function



[illegible]

The screenshot displays the AWS Lambda console for the 'dev-e-commerce-user-service' function. The 'Function overview' tab is active, showing the function's configuration. The function is triggered by an API Gateway, and its code is stored in an S3 bucket. The 'Code source' tab is selected, showing the function code in a code editor. The code defines an 'exports.handler' function that creates a new user in a DynamoDB database and returns a response.

```
exports.handler = async (event) => {
  body = JSON.stringify({
    count: result.count || 0
  });

  if (httpMethod === 'POST') {
    // Create new user
    const body = JSON.parse(event.body || '{}');
    const userId = Date.now().toString();

    const user = {
      id: userId,
      ...body,
      createdAt: new Date().toISOString()
    };

    await dynamoDb.put(new PutCommand({
      TableName: process.env.DDB_TABLE,
      Item: user
    }));
  }
}
```

AWS IAM: Roles

IAM > Roles

Identity and Access Management (IAM)

Dashboard

Access Management

Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests

Access reports

Access Analyzer
Resource analysis
Unusual access
Analysier settings
Credential report
Organization activity
Service control policies
Resource control policies

Related content

IAM Identity Center
AWS Organizations

Roles (17)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

| Role name | Trusted entities | Last activity |
|---|-------------------------------------|----------------|
| AmazonAPIGatewayPushToCloudWatchLogsRole | AWS Service: apigateway | 18 minutes ago |
| git-e-commerce-sts-oid2-getbotoresourcesrole-LO3HA6N7he5W | AWS Service: lambda | 2 days ago |
| AWSServiceRoleforAPIGateway | AWS Service: apigateway | - |
| AWSServiceRoleforEC2 | AWS Service: sts | 8 minutes ago |
| AWSServiceRoleforResourceExplorer | AWS Service: resource-explorer-2 | 6 minutes ago |
| AWSServiceRoleforSupport | AWS Service: support | 13 days ago |
| AWSServiceRoleforTrustedAdvisor | AWS Service: trustedadvisor | - |
| dev-aurora-enhanced-monitoring-role | AWS Service: monitoring | - |
| dev-e-commerce-apigateway-execution-role | AWS Service: apigateway | - |
| dev-e-commerce-authenticated-role | Identity Provider: cognito-identity | - |
| dev-e-commerce-cognito-authenticated-role | Identity Provider: cognito-identity | - |
| dev-e-commerce-lambda-execution-role | AWS Service: lambda | - |
| dev-e-commerce-lambda-role | AWS Service: lambda | 18 minutes ago |
| dev-firebase-delivery-role | AWS Service: firebase | 3 minutes ago |
| dev-glue-crawler-role | AWS Service: glue | 5 hours ago |
| ecommerce-application-role | AWS Service: lambda, and 3 more | - |
| ecommerce-devops-infrastructure-role | AWS Service: cloudformation, and 5 | - |

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

▼ Access reports

Access Analyzer

Resource analysis

Unused access

Analyzer settings

Credential report

Organization activity

dev-e-commerce-lambda-role

dev-e-commerce-lambda-role

Summary

Creation date
June 24, 2026, 00:30 (UTC+05:30)

Last activity
22 minutes ago

ARN
arn:aws:iam:759945100268:role/dev-e-commerce-lambda-role

Maximum session duration
1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (3)

You can attach up to 10 managed policies.

Filter by Type
All types

| Policy name | Type | Attached entities |
|------------------------------|-----------------|-------------------|
| AmazonDynamoDBFullAccess | AWS managed | 1 |
| AWSLambdaBasicExecutionRole | AWS managed | 3 |
| DynamoDBAndEventBridgeAccess | Customer inline | 0 |

AWS KMS: Keys

Key Management Service (KMS)

AWS managed keys

Customer managed keys

▼ Custom key stores

AWS CloudHSM key stores

External key stores

AWS managed keys (6)

Filter keys by properties or tags

| Aliases | Key ID | Status |
|--------------------|--------------------------------------|---------|
| aws/s3 | 0314eda5-6239-4687-84f4-5ff64faaa81e | Enabled |
| aws/rds | 2e92dd0d-2079-49ce-88d4-ed534e787012 | Enabled |
| aws/codecommit | b88c7958-d2ed-422c-824d-fae9134b1b1c | Enabled |
| aws/secretsmanager | ba0b1eea-146f-431a-a4d5-aab39fb7b29b | Enabled |
| aws/lambda | cb00ee4d-7b09-48cf-8f4b-02b75f55c74a | Enabled |
| aws/sns | d8e2cf7d-72be-40b2-82aa-093fb72d8b41 | Enabled |

Key policy

Cryptographic configuration

Key policy

```
1 {
2   "Version": "2012-10-17",
3   "Id": "auto-awslambda",
4   "Statement": [
5     {
6       "Sid": "Allow access through AWS Lambda for all principals in the account that are authorized to use AWS Lambda",
7       "Effect": "Allow",
8       "Principal": {
9         "AWS": "*"
10      },
11      "Action": [
12        "kms:Encrypt",
13        "kms:Decrypt",
14        "kms:ReEncrypt*",
15        "kms:GenerateDataKey*",
16        "kms:CreateGrant",
```

AWS CloudWatch: Log Management

CloudWatch

Log management

Log groups (16)

By default, we only load up to 10,000 log groups.

Filter log groups or try pattern search

Exact match

| Log group | Log class | Anomaly d... | Deletion pro... | Bearer token... | Data prote... | Sensitive ... | Retention | Metric filters |
|--|-----------|--------------|-----------------|-----------------|---------------|---------------|--------------|----------------|
| /aws/apigateway/welcome | Standard | Configure | Off | Off | - | - | Never expire | - |
| /aws/lambda/dev-e-commerce-analytics-service | Standard | Configure | Off | Off | - | - | Never expire | - |
| /aws/lambda/dev-e-commerce-inventory-service | Standard | Configure | Off | Off | - | - | Never expire | - |
| /aws/lambda/dev-e-commerce-notification-service | Standard | Configure | Off | Off | - | - | Never expire | - |
| /aws/lambda/dev-e-commerce-order-service | Standard | Configure | Off | Off | - | - | Never expire | - |
| /aws/lambda/dev-e-commerce-payment-service | Standard | Configure | Off | Off | - | - | Never expire | - |
| /aws/lambda/dev-e-commerce-product-service | Standard | Configure | Off | Off | - | - | Never expire | - |
| /aws/lambda/dev-e-commerce-search-service | Standard | Configure | Off | Off | - | - | Never expire | - |
| /aws/lambda/dev-e-commerce-user-service | Standard | Configure | Off | Off | - | - | Never expire | - |
| /aws/lambda/dev-get-root-resource | Standard | Configure | Off | Off | - | - | Never expire | - |
| /aws/rds/cluster/dev-e-commerce-aurora-cluster/error | Standard | Configure | Off | Off | - | - | 1 week | - |
| /aws/rds/cluster/dev-e-commerce-aurora-cluster/general | Standard | Configure | Off | Off | - | - | 3 days | - |
| /aws/rds/cluster/dev-e-commerce-aurora-cluster/slow... | Standard | Configure | Off | Off | - | - | 1 week | - |
| API-Gateway-Execution-Logs_4kljfdpxl/prod | Standard | Configure | Off | Off | - | - | Never expire | - |

Back to top

AWS Cognito: User Pools, Identity Pools and App Clients

Amazon Cognito

Current user pool

dev-e-commerce-users

Overview

Applications

User management

Authentication

Introducing Multi-Region replication

Get started

Overview: dev-e-commerce-users

Rename

Delete user pool

User pool information

User pool name

dev-e-commerce-users

User pool ID

ARN

Estimated number of users

2

OpenID Connect configuration URL

https://cognito-idp.us-east-1.amazonaws.com/us-east-1_4UEfsOvNej/well-known/openid-configuration

Token signing key URL

https://cognito-idp.us-east-1.amazonaws.com/us-east-1_4UEfsOvNej/well-known/jwks.json

Feature plan

Essentials

Created time

June 24, 2026 at 00:28 GMT+5:30

Last updated time

June 24, 2026 at 00:28 GMT+5:30

Regions

US East (N. Virginia) - Primary

Current Region

Create a replica user pool

Amazon Cognito

Current user pool

dev-e-commerce-users

Overview

Applications

App clients and analytics

App clients (1)

Delete

Create app client

Configure an app client. App clients are the user pool authentication resources attached to your app. Select an app client to configure the permitted authentication actions for an app.

Search app clients by name or ID

App client name

Client ID

dev-e-commerce-web-client

5m7aevm3b75dam2h8dm6jsor

Amazon Cognito

User pools

Identity pools

Identity pools (2)

Delete

Create identity pool

View and configure your identity pools. Identity pools enable you to create unique identities and assign permissions for users.

Search identity pools by name or ID

Identity pool name

Identity pool ID

Created time

Last updated time

dev_e-commerce_identity_pool

us-east-1:6f36a618-4ce7-480b-88fe-030c44e79d95

4 days ago

4 days ago

AWS SNS: Topics and Subscriptions

Amazon SNS

Dashboard

Topics

Subscriptions

Mobile

New Feature

Amazon SNS now supports High Throughput FIFO topics. Learn more

Topics (5)

Edit

Delete

Publish message

Create topic

Search

Name

Type

ARN

dev-notifications

Standard

arn:aws:sns:us-east-1:759945100268:dev-notifications

dev-order-events

Standard

arn:aws:sns:us-east-1:759945100268:dev-order-events

dev-payment-events

Standard

arn:aws:sns:us-east-1:759945100268:dev-payment-events

dev-product-events

Standard

arn:aws:sns:us-east-1:759945100268:dev-product-events

dev-user-events

Standard

arn:aws:sns:us-east-1:759945100268:dev-user-events

Amazon SNS

Dashboard

Topics

Subscriptions

Mobile

New Feature

Amazon SNS now supports High Throughput FIFO topics. Learn more

Subscriptions (4)

Edit

Delete

Request confirmation

Confirm subscription

Create subscription

Search

ID

Endpoint

Status

Protocol

Topic

137e01ee-0351-4c07-ac35-a0bc67155...

arn:aws:sqs:us-east-1:759945100268:de...

Confirmed

SQS

dev-order-events

c6110d42-1a37-49a3-95ec-e5771f36b...

arn:aws:sqs:us-east-1:759945100268:de...

Confirmed

SQS

dev-payment-events

74199f31-e9a1-4e6b-a364-4f67c3d97...

arn:aws:sqs:us-east-1:759945100268:de...

Confirmed

SQS

dev-product-events

ddc42ac5-66a1-40b7-a837-a31b09600...

arn:aws:sqs:us-east-1:759945100268:de...

Confirmed

SQS

dev-user-events

AWS EventBridge: Event Buses

Amazon EventBridge

Dashboard

Developer resources

Learn

Sandbox

Quick starts

Buses

Event buses

Rules Updated

Global endpoints

Archives

Replays

Pipes

Pipes

Scheduler

Scheduler

Event buses

Send events

Event buses receive events from a variety of sources and match them to rules in your account. Different types of event buses receive events from different sources, including AWS services in your account and other accounts, custom applications and services, and partner applications and services.

Enabling event discovery on an event bus will generate EventBridge Schemas for events on that bus. This may incur a cost (the first five million ingested events in each month is free).

Default event bus

Start discovery

Create archive

Send events

Edit

CloudFormation Template

Name

Amazon Resource Name (ARN)

Schema discovery

default

arn:aws:events:us-east-1:759945100268:event-bus/default

Not Initiated

Custom event bus (1)

Start discovery

Create archive

Send events

Edit

Delete

CloudFormation Template

Create event bus

Find custom event bus

Name

Amazon Resource Name (ARN)

Schema discovery

dev-ecommerce-events

arn:aws:events:us-east-1:759945100268:event-bus/dev-ecommerce-events

Not Initiated

Amazon EventBridge

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Buses

Event buses

Rules Updated

Global endpoints

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Pipes

Scheduler

Schedules

Schedule groups

Scheduled rules (legacy) Updated

Integration

Partner event sources

Rules

Encryption

Permissions

Monitoring

Logs

Archives

Tags

Rules on dev-ecommerce-events event bus (5)

Find rules

Any status

Delete

Enable

Edit

CloudFormation Template

Create rule

| | Name | Status | Type | Event bus | ARN | Description |
|--------------------------|---------------------------------|---------|----------|----------------------|---|--------------------------------------|
| ☐ | dev-ecommerce-notification-rule | Enabled | Standard | dev-ecommerce-events | arn:aws:events:us-east-1:759945100268:rule/dev-ecommerce-events/dev-ecommerce-notification-rule | Route events to notification service |
| ☐ | dev-order-created-rule | Enabled | Standard | dev-ecommerce-events | arn:aws:events:us-east-1:759945100268:rule/dev-ecommerce-events/dev-order-created-rule | - |
| ☐ | dev-payment-completed-rule | Enabled | Standard | dev-ecommerce-events | arn:aws:events:us-east-1:759945100268:rule/dev-ecommerce-events/dev-payment-completed-rule | - |
| ☐ | dev-product-created-rule | Enabled | Standard | dev-ecommerce-events | arn:aws:events:us-east-1:759945100268:rule/dev-ecommerce-events/dev-product-created-rule | - |
| ☐ | dev-user-registered-rule | Enabled | Standard | dev-ecommerce-events | arn:aws:events:us-east-1:759945100268:rule/dev-ecommerce-events/dev-user-registered-rule | - |

AWS Simple Queue Service: Queues

Queues (6)

Search queues by prefix

Edit

Delete

Send and receive messages

Actions

Create queue

| | Name | Type | Created | Messages available | Messages in flight | Encryption | Content-based deduplication |
|-----------------------|------------------------|----------|------------------------|--------------------|--------------------|--|-----------------------------|
| ☐ | dev-analytics-dlq | Standard | 2026-06-23T19:54+05:30 | 0 | 0 | AWS Key Management Service key (SSE-KMS) | - |
| ☐ | dev-analytics-queue | Standard | 2026-06-23T19:54+05:30 | 0 | 0 | AWS Key Management Service key (SSE-KMS) | - |
| ☐ | dev-notification-dlq | Standard | 2026-06-23T19:54+05:30 | 0 | 0 | AWS Key Management Service key (SSE-KMS) | - |
| ☐ | dev-notification-queue | Standard | 2026-06-23T19:54+05:30 | 0 | 0 | AWS Key Management Service key (SSE-KMS) | - |
| ☐ | dev-search-index-dlq | Standard | 2026-06-23T19:54+05:30 | 0 | 0 | AWS Key Management Service key (SSE-KMS) | - |
| ☐ | dev-search-index-queue | Standard | 2026-06-23T19:54+05:30 | 0 | 0 | AWS Key Management Service key (SSE-KMS) | - |

AWS Kinesis: Data Streams

Amazon Kinesis

Dashboard

Settings New

Data streams

Amazon Data Firehose New

Managed Apache Flink New

Resources

CloudFormation templates

AWS Glue Schema Registry

Amazon Kinesis Data Firehose is now called Amazon Data Firehose

Go to Amazon Data Firehose

dev-clickstream

Delete

Data stream summary

Status Active

Data retention period 1 day

ARN arn:aws:kinesis:us-east-1:759945100268:stream/dev-clickstream

Creation time June 24, 2026 at 00:06 GMT+5:30

Capacity mode Provisioned

Maximum record size 1024 KiB

Applications

Monitoring

Configuration

Enhanced fan-out (0)

Data viewer

Data analytics - new

Data stream sharing

EventBridge Pipes

Data stream capacity

Capacity mode Provisioned

Provisioned shards 2

Write capacity Maximum 2 MiB/second 2,000 records/second

Read capacity Maximum 4 MiB/second

Maximum record size

Maximum record size 1024 KiB

AWS Glue: Database, Crawler and Tables

AWS Glue

Getting started

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Data Catalog tables

Data connections

Workflows (orchestration)

Zero-ETL integrations

New

Data Catalog

Catalogs

Databases

Tables

Stream schema registries

Schemas

Connections

Find more

Job run monitoring

Data Catalog tables

Data connections

Workflows (orchestration)

Zero-ETL integrations

New

Data Catalog

Catalogs

Databases

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Stream schema registries

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Connections

Crawlers

Announcing enhancements to Glue statistics

We've added the option to schedule incremental stats generation, and stats for Iceberg tables. [Learn more](#)

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dev_ecommerce_analytics

Last updated (UTC)

June 27, 2026 at 07:34:31

Edit

Delete

Database properties

Name

dev_ecommerce_analytics

Description

Analytics database for e-commerce platform

Location

-

Created on (UTC)

June 23, 2026 at 18:36:34

Tables (1)

Last updated (UTC)

June 27, 2026 at 07:34:34

Delete

Add tables using crawler

Add table

View and manage all available tables.

Filter tables

☐

Name

☐

dev_clickstream_table

☐

Database

☐

dev_ecommerce_analytics

☐

Location

☐

s3://dev-e-commerce-data-l

☐

Classification

☐

Parquet

☐

Deprecated

☐

-

☐

View data

☐

Table data

☐

Data quality

☐

View data quality

☐

Column statistics

☐

View statistics

Tables (1)

Last updated (UTC)

June 27, 2026 at 07:34:35

Delete

Add tables using crawler

Add table

View and manage all available tables.

Choose catalog

759945100268

Choose database

dev_ecommerce_analytics

Filter tables

☐

Name

☐

dev_clickstream_table

☐

Database

☐

dev_ecommerce_analytics

☐

Location

☐

s3://dev-e-commerce-data-l

☐

Classification

☐

Parquet

☐

Deprecated

☐

-

☐

View data

☐

Table data

☐

Data quality

☐

View data quality

☐

Column statistics

☐

View statistics

AWS Glue

Getting started

ETL jobs

Visual ETL

Notebooks

Job run monitoring

Data Catalog tables

Data connections

Workflows (orchestration)

Zero-ETL integrations

New

Data Catalog

Catalogs

Databases

Tables

Stream schema registries

Schemas

Connections

Crawlers

Classifiers

Catalog settings

Data Integration and ETL

Legacy pages

What's New

Documentation

AWS Marketplace

Enable compact mode

Announcing new optimization features for Apache Iceberg tables

Optimize storage for Apache Iceberg tables with automatic snapshot retention and orphan file deletion. [Learn more](#)

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dev-clickstream-crawler

Last updated (UTC)

June 27, 2026 at 07:35:00

Run crawler

Edit

Delete

Crawler properties

Name

dev-clickstream-crawler

IAM role

dev-glue-crawler-role

Database

dev_ecommerce_analytics

State

READY

Description

-

Security configuration

-

Lake Formation configuration

-

Table prefix

-

Maximum table threshold

-

Advanced settings

Crawler runs

Schedule

Data sources

Classifiers

Tags

Crawler runs (4)

The list of crawler runs for this crawler.

Stop run

View CloudWatch logs

View run details

Filter data

Filter by a date and time range

☐

Start time (UTC)

☐

End time (UTC)

☐

Current/last duration

☐

Status

☐

DPU hours

☐

Table changes

☐

June 27, 2026 at 02:04:04

☐

June 26, 2026 at 02:03:17

☐

June 25, 2026 at 02:04:02

☐

June 24, 2026 at 02:03:09

☐

June 27, 2026 at 02:04:57

☐

June 26, 2026 at 02:04:04

☐

June 25, 2026 at 02:04:47

☐

June 24, 2026 at 02:05:26

☐

52 s

☐

46 s

☐

44 s

☐

02 min 16 s

☐

Completed

☐

Completed

☐

Completed

☐

Completed

☐

0.037

☐

0.128

☐

0.125

☐

0.039

☐

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AWS Kinesis Firehose: Firehose Streams

Amazon Data Firehose

Firehose streams

Resources

What's new

Developer guide

API reference

Firehose stream details

Status

Active

Destination

Amazon S3

Data transformation

Not enabled

Creation time

June 24, 2026 at 00:07 GMT+5:30

Source

Amazon Kinesis Data Streams

ARN

arn:aws:firehose:us-east-1:759945100268:deliverystream/dev-clickstream-firehose

Dynamic partitioning

Not enabled

Test with demo data

Info

Ingest simulated data to test the configuration of your Firehose stream. Standard Amazon Data Firehose charges apply.

Monitoring

Configuration

Source settings

Kinesis data stream

dev-clickstream

AWS S3: Bucket

Buckets

General purpose buckets All AWS Regions Directory buckets

General purpose buckets (2) Info

Buckets are containers for data stored in S3.

Find buckets by name

| | Name | AWS Region | Creation date |
|-----------------------|------------------------------------|---------------------------------|-------------------------------------|
| ☐ | cf-templates-zr4gdtkl13x-us-east-1 | US East (N. Virginia) us-east-1 | June 23, 2026, 14:47:39 (UTC+05:30) |
| ☐ | your-e-commerce-frontend-bucket | US East (N. Virginia) us-east-1 | June 26, 2026, 12:43:51 (UTC+05:30) |

Account snapshot Info

Updated daily

Storage Lens provides visibility into storage usage and activity trends.

[View dashboard](#)

External access summary Info

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

Amazon S3 Buckets your-e-commerce-frontend-bucket

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block all public access
Off

Individual Block Public Access settings for this bucket

[Edit](#)

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

[Edit](#)

[Delete](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3::your-e-commerce-frontend-bucket/*"
    }
  ]
}
```

[Copy](#)

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

[Edit](#)

We recommend using AWS Amplify Hosting for static website hosting

Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about [Amplify Hosting](#) or [View your existing Amplify apps](#)

[Create Amplify app](#)

S3 static website hosting

Enabled

Hosting type

Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://your-e-commerce-frontend-bucket.s3-website-us-east-1.amazonaws.com>

Amazon S3 Buckets your-e-commerce-frontend-bucket

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems

Access management and security

your-e-commerce-frontend-bucket Info

Objects Metadata Properties Permissions Metrics Management File systems - new Access Points

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

| | Name | Type | Last modified | Size | Storage class |
|--------------------------|----------------------------|------|-------------------------------------|---------|---------------|
| ☐ | index.html | html | June 27, 2026, 00:03:53 (UTC+05:30) | 10.7 KB | Standard |

AWS Aurora and RDS: Databases along with secrets in Secret Manager and Security Groups in EC2

Aurora and RDS Databases

Aurora and RDS

Dashboard

Databases

Query editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Databases (5)

Filter by databases

Group resources

[Refresh](#)

[Reset](#)

[Modify](#)

[Actions](#)

[Create database](#)

| | DB identifier | Status | Role | Engine | Upgrade rollout order | Region | Size | Recommendations | CPU |
|-----------------------|-------------------------------|-----------|---------------|--------------|-----------------------|------------|-------------------------|-----------------|-------|
| ☐ | dev-e-commerce-aurora-cluster | Available | Regional c... | Aurora My... | SECOND | us-east-1 | 2 instances | 5 Informational | - |
| ☐ | dev-e-commerce-aurora-reader | Available | Writer ins... | Aurora My... | SECOND | us-east-1d | Aurora serverless (0... | 2 Informational | 2.83% |
| ☐ | dev-e-commerce-aurora-writer | Available | Reader ins... | Aurora My... | SECOND | us-east-1c | Aurora serverless (0... | 2 Informational | 2.57% |

Secrets

Filter secrets by name, description, tag key, tag value, owning service or primary Region

| Secret name | Description | Last retrieved (UTC) | Created on (UTC) |
|----------------------------------|--|----------------------|---------------------------|
| dev/e-commerce/aurora/connection | Aurora MySQL connection details for e-commerce application | - | June 23, 2026 at 19:55:01 |

EC2 > Security Groups

Find security groups by attribute or tag

sg-083414d1f8e978c8b Clear filters

| Name | Security group ID | Security group name | VPC ID | Description | Owner |
|---------------|----------------------|---|-----------------------|---|--------------|
| dev-aurora-sg | sg-083414d1f8e978c8b | aurora-stack-ecom02-AuroraSecurityGr... | vpc-0a07fdead56748f13 | Security group for Aurora MySQL cluster | 759945100268 |

Testing

Not secure your-e-commerce-frontend-bucket.s3-website-us-east-1.amazonaws.com

System Status: OPERATIONAL
All services running optimally • Last updated: 6/27/2026, 2:14:23 PM

Backend Services

- ✓ Product Management Service
- ✓ Order Processing Service
- ✓ User Management Service
- ✓ Payment Processing Service
- ✓ Inventory Management Service
- ✓ Search & Analytics Service
- ✓ Notification Service
- ✓ API Gateway (Regional)

Data Layer

- ✓ Aurora MySQL Serverless
- ✓ DynamoDB Tables (10)
- ✓ S3 Data Lake
- ✓ EventBridge Event Bus
- ✓ SQS Message Queues (6)
- ✓ SNS Topics (5)

Security & Auth

- ✓ Cognito User Pools (2)
- ✓ IAM Roles & Policies
- ✓ KMS Encryption
- ✓ VPC Security Groups
- ✓ API Gateway Throttling
- ✓ CloudWatch Monitoring

API Gateway Endpoints

Base URL: https://4ki1jfdpxrl.execute-api.us-east-1.amazonaws.com/prod

17 API Resources

10K Requests/sec

99.9% Uptime

< 100ms Avg Latency

Test Products API Test Orders API Test Users API Test Inventory API View API Console

System Status: OPERATIONAL
All services running optimally • Last updated: 6/27/2026, 1:20:22 PM

Backend Services

- ✓ Product Management Service
- ✓ Order Processing Service
- ✓ User Management Service
- ✓ Payment Processing Service
- ✓ Inventory Management Service
- ✓ Search & Analytics Service
- ✓ Notification Service
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Data Layer

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- ✓ IAM Roles & Policies
- ✓ KMS Encryption
- ✓ VPC Security Groups
- ✓ API Gateway Throttling
- ✓ CloudWatch Monitoring

API Gateway Endpoints

Base URL: https://4ki1jfdpxrl.execute-api.us-east-1.amazonaws.com/prod

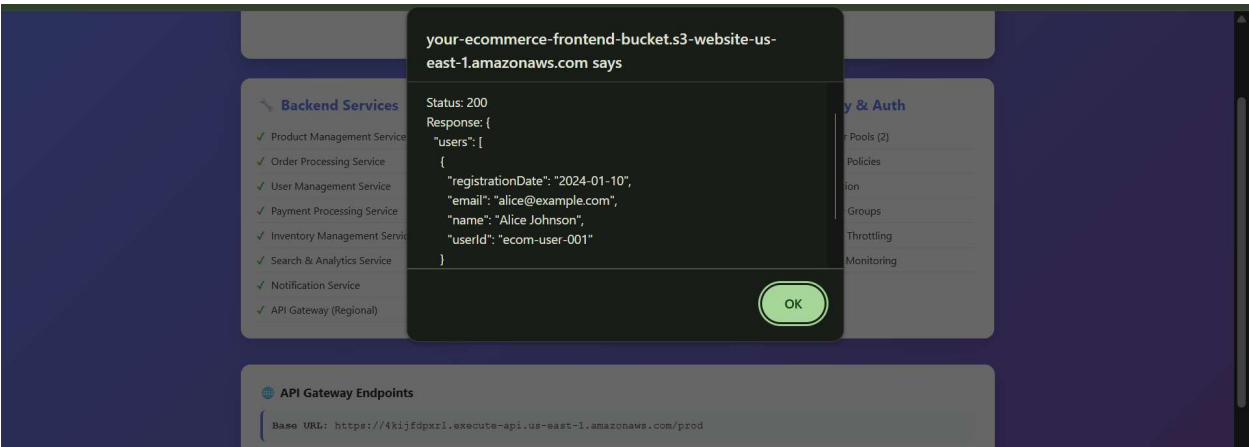
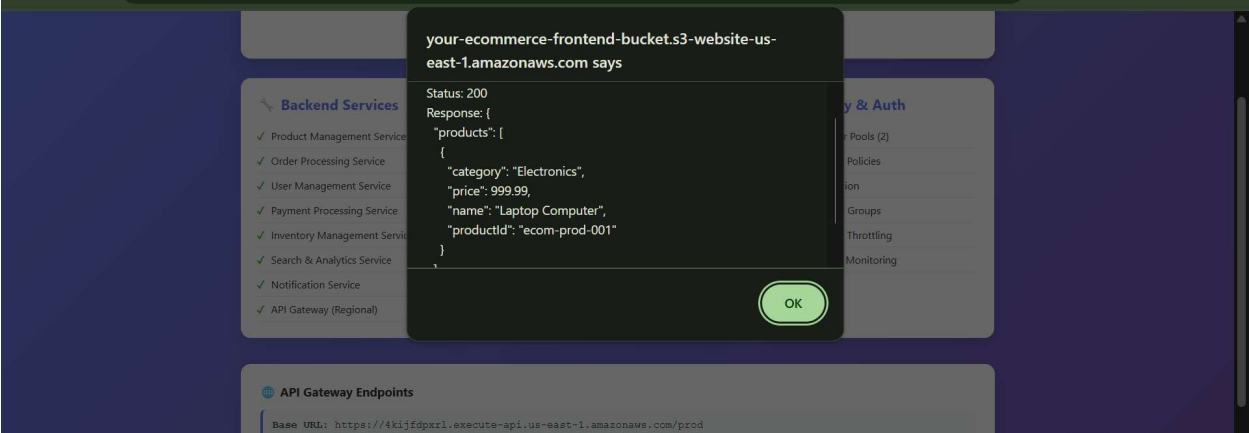
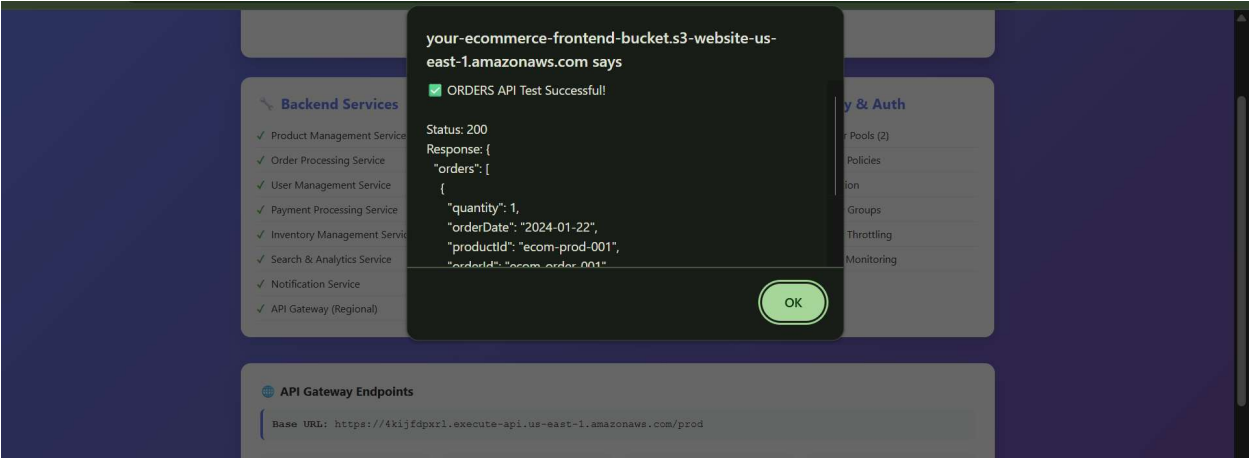
17 API Resources

10K Requests/sec

99.9% Uptime

< 100ms Avg Latency

Test Products API Test Orders API Test Users API Test Inventory API View API Console



EXPLORER

NEW FOLDER

dist

node_modules

public

src

api

assets

components

pages

AnalyticsPage.tsx

CartPage.tsx

HomePage.tsx

InventoryPage.tsx

OrdersPage.tsx

ProductDetailP...

ProfilePage.tsx

store

types

App.css

App.tsx

index.css

main.tsx

.gitignore

.oxlintrc.json

index.html

TIMELINE

CartPage.tsx X

src > pages > CartPage.tsx > ...
1 import { useMemo, useState } from "react";
2 import { useNavigate } from "react-router-dom";
3 import { createOrder } from "../api/orders";
4 import { createPayment } from "../api/payments";
5 import { useCartStore } from "../store/cartStore";
6 import type { CartItem } from "../types";
7 import { useAuthStore } from "../store/authStore";
8
9 const CartPage = () => {
10 const items = useCartStore((state) => state.items);
11 const clearCart = useCartStore((state) => state.clearCart);
12 const user = useAuthStore((state) => state.user);
13 const navigate = useNavigate();
14 const [message, setMessage] = useState("");
15
16 const total = useMemo(
17 () => items.reduce((sum, item) => sum + item.price * item.quantity, 0),
18 [items],
19);
20
21 const handleCheckout = async () => {
22 if (!user) {
23 setMessage("Please sign in to continue.");
24 return;
25 }
26
27 try {
28 const order = await createOrder({
29 userId: user.id,

